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ALGEBRAIC STACKS

In classical algebraic geometry, we have the following definition of algebraic stacks.

Definition: A functor

$$X : \text{Ring}^{\text{static}} \longrightarrow \text{Gpd}$$

is an (fppf-) algebraic stack if it satisfies:

① X satisfies fppf-descent, i.e., for any fppf cover $\text{Spec } B \xrightarrow{\varphi} \text{Spec } A$, and any pair (E, σ) where $E \in X(B)$ and

$$\sigma : \text{pr}_1^* E \xrightarrow{\sim} \text{pr}_2^* E$$

over $B \otimes_A B$, such that over $B \otimes_A B \otimes_A B$

the following diagram commutes

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$$\begin{array}{ccc}
 \text{pr}_{12}^* \text{pr}_1^* E & \xrightarrow{\text{pr}_{12}^* \sigma} & \text{pr}_{12}^* \text{pr}_2^* E \quad \quad \quad \text{pr}_{23}^* \text{pr}_1^* E \\
 \parallel & & \downarrow \text{pr}_{23}^* \sigma \\
 \text{pr}_{13}^* \text{pr}_1^* E & \xrightarrow{\text{pr}_{13}^* \sigma} & \text{pr}_{13}^* \text{pr}_2^* E \quad \quad \quad \text{pr}_{23}^* \text{pr}_2^* E, \\
 & & | \\
 & & \text{pr}_{23}^* \text{pr}_1^* E
 \end{array}$$

we can define a category $X(\text{Spec } B \rightarrow \text{Spec } A)$ of such pairs. Then, the functor

$$X(\text{Spec } A) \xrightarrow{\varphi^*} X(\text{Spec } B \rightarrow \text{Spec } A)$$

is an equivalence.

② The diagonal map

$$\Delta: X \longrightarrow X \times X$$

is representable, i.e., for any $A \in \text{Ring}^{\text{static}}$,

the fiber product

$$\begin{array}{ccc}
 X & \longrightarrow & X \times X \\
 \uparrow & \lrcorner & \uparrow \\
 \mathcal{E} & \dashrightarrow & \text{Spec } A
 \end{array}$$

is representable by an algebraic space.

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③ There is a smooth epimorphism $S \rightarrow X$,
for some scheme S .

We wish define something similar in the
derived setting.

Notation:

$\text{Aff} := \text{Ring}^{\text{op}}$ category of affine schemes.

And, for $R \in \text{Ring}$, we denote by
 $\text{Spec } R \in \text{Aff}$ the corresponding object, and
denote the underlying topological space by $|\text{Spec } R|$.

Definition: The category of algebraic prestacks
is the full subcategory

$$\text{PreStk} \subseteq \text{Fun}(\text{Ring}, \text{Ani})$$

See Remark 5.2 in
Camargo's notes for
more details

generated under (small) colimits by Aff via the Yoneda
Embedding

Suppose $T \in \{\text{Zariski}, \text{étale}, \text{fppf}, \text{fpqc}\}$ is a topology on Aff (more generally, T is an "accessible" Grothendieck topology).

Moreover, for an uncountable regular cardinal $\kappa \geq \omega_1$, let Aff^κ be the category of κ -light affine schemes.

Definition: We define the category of κ -light algebraic stacks to be

$$\text{AlgStk}_T^{\kappa\text{-light}} \subseteq \text{Fun}(\text{Aff}^{\kappa, \text{op}}, \text{Ani})$$

such that for any $X \in \text{Aff}^\kappa$, and covering sieve $\mathcal{C}_{/X}^{(o)} \in \text{Sieve}_T(X)$

$$F(X) \longrightarrow \lim_{Y \in \mathcal{C}_{/X}^{(o)}} F(Y)$$

is an equivalence.

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Now, if $\kappa' \geq \kappa$ is another regular cardinal, then the left Kan extension

$$\text{Fun}(\text{Aff}^{\kappa, \text{op}}, \text{Ani}) \subseteq \text{Fun}(\text{Aff}^{\kappa', \text{op}}, \text{Ani})$$

$$\begin{array}{ccc} & \text{Aff}^{\kappa', \text{op}} & \\ \nearrow & \uparrow \eta & \searrow \tilde{\tau}_{\kappa'}(\varepsilon) \\ \text{Aff}^{\kappa, \text{op}} & \xrightarrow{\mathbb{F}} & \text{Ani} \end{array}$$

defines $\text{AlgStk}_{\mathbb{T}}^{\kappa\text{-light}} \xrightarrow{T_{\kappa \rightarrow \kappa'}} \text{AlgStk}_{\mathbb{T}}^{\kappa'\text{-light}},$

such that $T_{\kappa \rightarrow \kappa'}$ is a fully-faithful, left exact colimit preserving functor.

Then, we define the category of algebraic stacks as

$$\text{AlgStk}_{\mathbb{T}} := \text{colim}_{\kappa} \text{AlgStk}_{\mathbb{T}}^{\kappa\text{-light}}$$

along $T_{\kappa \rightarrow \kappa'}$ for all uncountable regular cardinals κ

Example : Classical algebraic stacks.

⑥

Let $\text{Aff}_{\leq n} \subseteq \text{Aff}$ be the full subcategory of n -truncated affine schemes, and

$T \in \{\text{Zariski}, \text{étale}, \text{fppf}, \text{fpqc}\}$ as before.

Let $\text{Sh}_T(\text{Aff}_{\leq n}, \text{Ani})$ be the category of sheaves on $\text{Aff}_{\leq n}$ (defined in a manner similar to AlgStk_T using regular cardinals).

Then, we have a fully faithful embedding of sites $\text{Aff}_{\leq n, T} \subseteq \text{Aff}$ which induces a left exact, colimit preserving functor

$$F_n : \text{Sh}_T(\text{Aff}_{\leq n}, \text{Ani}) \longrightarrow \text{AlgStk}_T.$$

An object $X \in \text{AlgStk}_T$ is called n -classical if it is in the essential image of F_n .

Further, it is called classical if it is ⑦
0 - classical.

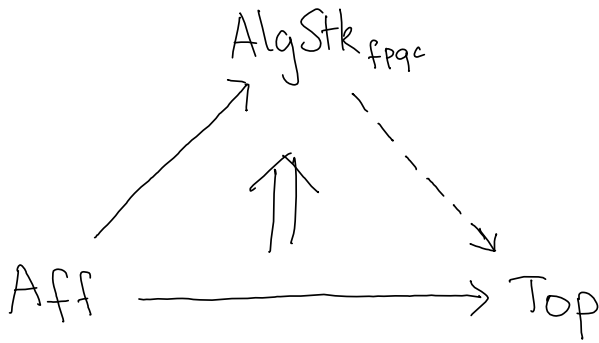
Lemma: Let $X = \text{Spec } R$ be an affine
scheme viewed as an algebraic prestack
 $X \in \text{PreStk}$ via the Yoneda embedding.
The X is an fpqc-stack.

Slogan: The fpqc topology is subcanonical

Consider the Zariski spectrum functor
defined previously,

$$|\cdot| : \text{Aff} \longrightarrow \text{Top}.$$

Then, (because our categories are nice) we
can define a left Kan extension



And, we continue to denote and call the functor

$$|\cdot| : \text{AlgStk}_{\text{fpqc}} \longrightarrow \text{Top}$$

the underlying Zariski spectrum functor.

Now, suppose $T \in \{\text{Zariski}, \text{etale}, \text{fppf}, \text{fpqc}\}$ and

$Y \in \text{AlgStk}_T$. If $|Y|$ is the underlying

Zariski spectrum and $|U| \subseteq |Y|$ is an

open subspace, we define the the

substack, whose $\text{Spec } R$ points are

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$$U(R) = \begin{cases} Y(R) & \text{if } |\mathrm{Spec} R| \rightarrow |Y| \text{ factors through } U \\ \emptyset & \text{otherwise.} \end{cases}$$

Definition: A scheme X is a Zariski algebraic stack, i.e., $X \in \mathrm{AlgStk}_{\mathrm{zar}}$, such that $|X|$ admits an open cover $\{|U_i|\}_i$ such that U_i 's are affine schemes. We shall denote by $\mathrm{Sch} \subseteq \mathrm{AlgStk}_{\mathrm{zar}}$ the full subcategory of schemes.

Proposition: Let $X \in \mathrm{Sch}$ be a scheme. Then X is a sheaf for the fpqc topology. In particular, we have a fully faithful embedding $\mathrm{Sch} \subseteq \mathrm{AlgStk}_{\mathrm{fpqc}}$.

Consider the functor of derived categories of modules.

$$D : \text{Aff}^{\text{op}} \longrightarrow \text{CAlg}(\text{Cat}),$$

where $D(X)$ is a symmetric monoidal category for $X \in \text{Aff}$

Definition: We define the functor of quasicoherent sheaves on $\text{AlgStk}_{\text{fqc}}$ to be the right Kan extension of D

$$\begin{array}{ccc} & \text{AlgStk}_{\text{fqc}}^{\text{op}} & \\ \nearrow & \Downarrow \varepsilon & \searrow D \\ \text{Aff}^{\text{op}} & \xrightarrow{\quad} & \text{CAlg}(\text{Cat}) \end{array}$$

Given a morphism $f: Y \rightarrow X$ of algebraic stack, we let $f^* = D(f): D(X) \rightarrow D(Y)$

be the pullback functor (which is symmetric 11
monoidal), and let $f_*: \mathcal{D}(Y) \rightarrow \mathcal{D}(X)$
be its ^{right} adjoint (the relative cohomology
functor).

Further, we denote by $\otimes_x$ the tensor
product of $\mathcal{D}(X)$.

Further, we let $\underline{\text{Hom}}(\cdot, \cdot)$ be the
internal Hom functor of $\mathcal{D}(X)$ such
that

$$-\otimes_x M \rightarrow \underline{\text{Hom}}(M, -).$$

Proposition: The functor $\mathcal{D}: \text{Aff}^{\text{op}} \rightarrow \text{Cat}(\text{Alg})$ is
a sheaf for the fpqc topology. In particular
its right Kan extension

$$\mathcal{D}: \text{AlgStk}^{\text{op}} \rightarrow \text{Cat}(\text{Alg})$$

preserves limits.

Let $X \in \text{AlgStk}_{\text{fpqc}}$.

We define $D_{\geq 0}(X) \subseteq D(X)$ the full subcategory of those quasi-coherent sheaves $M \in D(X)$ such that for any map $f: \text{Spec } R \rightarrow X$, $f^* M \in D_{\geq 0}(R)$.

Now, let $A \in \text{Ring}$. Recall, that we have an adjunction

$$\text{Sym}_A^\bullet : D_{\geq 0}(A) \rightleftarrows \text{Ring}_{A/} : G(A)$$

Lemma: The functor

$$\text{Ring}_{(-)/} : \text{Aff}^{\text{op}} \longrightarrow \text{Cat}$$

sending $\text{Spec } R$ to $\text{Ring}_{R/}$ satisfies fpqc descent. Moreover, the natural transformation

$$G(-) : \text{Ring}_{(-)/} \longrightarrow D_{\geq 0}(-)$$

has a left adjoint given by the symmetric powers functor.

$$\text{Sym}_{(-)}^{\bullet} : D_{\geq}(-) \longrightarrow \text{Ring}_{(-)/}.$$

Definition: The functor of relative animated algebras over fpqc algebraic stacks is defined to be the right Kan extension of the $\text{Ring}_{(-)}$ functor.

$$\begin{array}{ccc}
 & \text{AlgStk}_{\text{fpqc}}^{\text{op}} & \\
 \nearrow & \Downarrow \varepsilon & \searrow \text{Ring}_{(-)} \\
 \text{Aff}^{\text{op}} & \xrightarrow{\quad} & \text{Cat}
 \end{array}$$

The forgetful natural transformation

$G(-) : \text{Ring}_{(-)} \longrightarrow \mathcal{D}_{\geq 0}(-)$ of functors from

$\text{AffStk}_{\text{fpqc}}^{\text{op}} \longrightarrow \text{Cat}$ is obtained as the right

Kan extension of $G(-)$ for Affine schemes.

Moreover, it has a left adjoint given by the right Kan extension of the symmetric power functor

$$\mathrm{Sym}_{(-)}^{\bullet} : D_{\geq 0}(-) \longrightarrow \mathrm{Ring}(-)$$

Definition: Suppose $X \in \mathrm{AlgStk}_{\mathrm{fpqc}}$ and $\mathcal{A} \in \mathrm{Ring}_X$. Then, the relative spectrum of $\mathcal{A}$ over X , denoted by $\mathrm{Spec}_X \mathcal{A}$ is the fpqc stack in $\mathrm{AlgStk}_{\mathrm{fpqc}}/X$ whose values at $f: \mathrm{Spec} R \rightarrow X$ are given by.

$$(\mathrm{Spec}_X \mathcal{A})(R) = \mathrm{Map}_{\mathrm{Ring}_R}(f^* \mathcal{A}, R)$$